Ojasvi Shrivastava

B.A. Data Science · UC Berkeley · Class of 2027

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SUMMARY

UC Berkeley Data Science student building production ready tools that move probabilistic modeling, spatial ML, and Monte Carlo simulation out of notebooks and into interactive applications. Seeking a Summer 2026 data science or ML internship.

EDUCATION

University of California, Berkeley — *Berkeley, CA*
B.A., Data Science — *Expected May 2027*
Relevant Coursework: Probability, Linear Algebra, Multivariable Calculus, Algorithms & Data Structures, Machine Learning, Artificial Intelligence

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, TypeScript, SQL
- **ML / Data:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Statsmodels, Monte Carlo / GBM, K-Means, Multivariate Regression, Spatial Econometrics, GeoPandas, Matplotlib, Plotly, Tableau
- **Web Development:** Next.js, React, TypeScript, Tailwind CSS, Recharts, Framer Motion, Vercel
- **Tools:** Git, Jupyter, Linux, ML Pipelines, Vectorization & Numerical Optimization, Algorithm Design

EXPERIENCE

Private Mathematics Tutor — *2024 – 2025*

- Identified binding conceptual gaps for **8+ students** across calculus, linear algebra, and statistics; re-sequenced material to address weak links first, producing an **average 60% improvement in exam scores**.
- Tailored instruction style per student visual derivations, worked examples, or Socratic dialogue and built reusable problem sets refined across sessions.
- Designed first principles frameworks enabling students to reconstruct any result from definitions on exam day, eliminating dependence on rote formula recall.

PROJECTS

FinSight — *Full-Stack Wealth & Retirement Simulator* — *2025 – Present*
TypeScript, Next.js, Monte Carlo / GBM, localStorage — *Live Site* *GitHub*

- Built a vectorized stochastic engine running **3,000-trial Geometric Brownian Motion** simulations with a seeded PRNG for deterministic P10/P50/P90 retirement forecasting; a What-If slider quantifies the real-time effect of an extra $200/month over 25 years.
- Developed a rule based **AI insight engine** that reads the user's actual savings rate, portfolio P&L, and emergency fund gap to generate specific, numbered recommendations instead of generic tips.
- Delivered the entire product with no backend: a cash flow tracker, emergency fund ring chart, live market sentiment gauge, and a **six-article financial education curriculum** with quizzes all via localStorage, no server or account required.

GIStice League — *Spatial Econometrics for Food-Access Inequity* — *Berkeley Open Project*
Machine Learning Team | Python, PyTorch, Scikit-learn, GeoPandas — *Live Site* *GitHub*

- Engineered K-Means spatial cluster identifiers and neighborhood grocery density kernels into a PyTorch multivariate regression model, achieving an **18% R^2 improvement** over distance only baselines across **10,000+ U.S. Census tracts**.
- Built a reproducible pipeline rendering interactive **Leaflet + GeoPandas choropleths** with 20+ live ML features and real-time R^2 / RMSE updates; community advocates adopted the tool for targeted food infrastructure policy decisions.